

VIJAYSINGH PUWAR

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PROFILE

Engineer working across two halves of the same stack: the network and systems infrastructure underneath, and the AI-powered applications running on top. Supports enterprise routing, switching, and zero-trust access for a transit operating agency, and separately designs, ships, and operates production platforms with paying users: multi-provider LLM pipelines, REST backends, and relational data layers, backed by **1,280+ automated tests**. Cisco CCNA, Security+ and CySA+ certified; CCNP Enterprise in progress. M.S. Cybersecurity, Pace University, GPA 3.92, December 2026.

EXPERIENCE

MTA STATEN ISLAND RAILWAY (SIRTOA) | June 2026 – Present

New York City

IT Emerging Talent Intern

- Configure **Cisco Catalyst** switching (VLANs, access and trunk ports) and support **OSPF and BGP routing on production network devices**.
- Administer **zero-trust and identity access**: **Zscaler ZPA** application groups, **DUO** multi-factor enrollment, **Active Directory** groups, and provisioning and revocation as staff join, move, and leave.
- Own incidents end to end in **ServiceNow** from triage through closure, maintain a daily System Verification Log, and deploy endpoint hardware at remote facilities.
- Designed a **Power Automate and SharePoint** intake application and **root-caused a trigger-versus-Compose defect** in testing; deployed a **Python NTP** clock-correction service for a Rail Control Center workstation.

R.S. INFOTECH | February 2023 – August 2024

Vadodara, India

System Engineer

- Configured **production firewall rules** and supported **IDS/IPS**, reviewing traffic and system logs to remediate misconfigurations and suspicious activity.
- Hardened **150+ Windows and Linux production endpoints** (baselines, patching, post-change validation) with **Windows Server**, **Active Directory**, and **Group Policy** administration.
- Built **automation in Python, PowerShell, and Bash** for log analysis, configuration-compliance checking, and inventory; investigated failed-authentication patterns with **Splunk SPL**.

PROJECTS

Recap Verse | AI engineering: Python, FastAPI, multi-provider LLMs, MongoDB, Cloud Run

[recapverse.com](#)

- Operate a paying multi-tenant platform on a **multi-provider LLM pipeline** (Claude Opus and Sonnet, GPT, Gemini) with bring-your-own-key isolation and **AES-GCM encrypted** per-user credentials, on a **19-router FastAPI backend** deployed to Google Cloud Run across **353 deployments**, with **900+ tests** run before every release.
- Architected the pipeline as **vision observation**, then a **deterministic five-stage resolver**, then **narration with Chain-of-Density rolling summaries**, then **deterministic post-processing**, where the design judgment is which stages a model may own and which must not be probabilistic.
- Instrumented it for measurement: an **evaluation harness**, **A/B feature flags** pitting a pipeline stage against its predecessor, reservoir-sampled model observations for telemetry replay, and capture of human edits against AI output.

Web Audio TV | full stack: TypeScript, React, PostgreSQL, PL/pgSQL, Deno

[webaudiotv.com](#)

- Own the data layer behind a live platform serving **843 users**: 40+ tables, **52 SQL functions**, **11 triggers**, **93 versioned migrations**, and 190+ row-level access policies across 24 edge functions.
- Built an **append-only rewards ledger** enforced by database trigger, **Stripe and PayPal billing** with webhook verification and scheduled reconciliation, and an **abuse-detection console** with named heuristics and behavioral baselining. **389 tests** across 39 files.

Networking & Security Labs | Cisco, Linux, Nmap NSE

[github.com/VijaysinghPuwar](#)

- **Dual-ISP assessment and remediation**: diagnosed a split-brain network with **overlapping RFC1918 subnets** via multi-vantage-point discovery, remediated it, and documented the result with topology diagrams and a written report.
- **Rapid-PVST campus lab** (per-VLAN root load balancing, PortFast, BPDU Guard, Root Guard), **iptables hardening** with default-deny and brute-force rate limiting under CI, and a custom **Nmap NSE** script checking HSTS, CSP, and X-Frame-Options.

EDUCATION

PACE UNIVERSITY | Seidenberg School of Computer Science and Information Systems

New York, NY

M.S. Cybersecurity, GPA 3.92/4.0 | Network Security & Defense, Ethical Hacking, Algorithms, Data Science

December 2026

G H PATEL COLLEGE OF ENGINEERING AND TECHNOLOGY

Anand, India

B.E. Mechanical Engineering

January 2024

SKILLS & CERTIFICATIONS

Certifications: Cisco CCNA, CompTIA Security+, CompTIA CySA+, Google AI Essentials; Cisco CCNP Enterprise (in progress); TryHackMe, top 7% globally, 69 rooms

Networking: Routing and switching, OSPF, BGP, VLANs and trunking, Rapid-PVST, TCP/IP, DNS, NTP, HTTP/TLS, subnetting, Cisco IOS, firewalls, IDS/IPS, VPN, zero-trust access

AI Engineering: Multi-provider LLM integration (Anthropic, OpenAI, Google), prompt design, deterministic stages around model output, evaluation harnesses, A/B feature flags, telemetry sampling

Programming: Python, TypeScript, JavaScript, SQL, PL/pgSQL, PowerShell, Bash, Lua

Software Engineering: FastAPI, REST API design, Next.js, React, Pydantic and Zod validation, PostgreSQL, MongoDB, Redis job queues, Docker, Cloud Run, Vercel, AWS, GitHub Actions CI, Windows Server, Active Directory, Linux, ServiceNow

Security: Authorization design, least privilege, secure architecture review, input validation, SSRF defense, secrets management, Nmap NSE